

Cyclic-Rank Obstructions and Growing-Horizon Stein Certificates

Minimal Gramians, Conic No-Go Witnesses, and a Viable Block Route

Liang Wang

July 2026

Abstract

The preceding two-pole Hardy reduction converted the remaining small-noise range-resolvent gate for the folded-Gaussian quadratic operator into positive Stein inequalities. We determine what an exact positive certificate can and cannot look like. For a stable scaled bulk matrix $A = N/r$, a directional source X , and

$$G = \sum_{m \geq 0} A^m X X^* (A^*)^m, \quad G - A G A^* = X X^*,$$

we prove that G is the minimal positive Stein supersolution and that

$$\text{Ran } G = \text{span}\{A^m \text{Ran } X : m \geq 0\}.$$

Every positive supersolution has an A -invariant range containing this cyclic subspace, hence rank at least $\text{rank } G$. Fixed-rank exact Lyapunov factors are therefore impossible whenever the directional cyclic dimension grows. If $H = Z Z^* + \alpha I \geq G$ and $\text{rank } Z \leq k$, then necessarily $\alpha \geq \lambda_{k+1}(G)$.

The obstruction does not close the Stein route. With

$$S_M = \sum_{j=0}^{M-1} A^j X X^* (A^*)^j,$$

the block inequality $H - A^M H (A^*)^M \geq S_M$ already implies $H \geq G$. If $q_M = \|A^M\| < 1$, then

$$H_M = S_M + \alpha_M I, \quad \alpha_M = \frac{\|A^M S_M (A^*)^M\|}{1 - q_M^2}$$

is an explicit block supersolution. Thus the fixed-step global-contraction no-go of the preceding paper is compatible with a horizon increasing with noise resolution. We also give an anisotropic residual completion and conic dual witnesses that rule out whole metric ansatz classes.

A five-scale binary64 dense audit, with $N\sigma = 5.12$, $r = 0.85$, and $N = 32, \dots, 512$, solves both directional Gramians. The left and right energies remain between 0.904–1.468 and 1.003–1.760, while the ranks carrying 99% of the Gramian trace grow from 5 to 69 and from 5 to 64. At relative singular threshold 10^{-8} , the left cyclic rank grows from 22 to 322, about $0.63N$ at the finest scale. Scalar identity metrics have rank-one dual obstructions at the four finest levels, and the diagonal extracted from the exact Gramian fails at all five. In contrast, block horizons 4, 8, 16, 24, 32 give energy uppers within 6.3×10^{-4} relative of the dense Gramian energies. These are floating diagnostics, not a dyadically uniform theorem. No arithmetic trace formula, zeta-zero identity, self-adjoint realization, or Riemann-hypothesis conclusion is claimed.

Keywords: Stein inequality; controllability Gramian; cyclic subspace; Lyapunov rank; block certificate; conic duality; small-noise transfer operator; Hardy energy.

MSC 2020: 47A10; 93B05; 93B36; 15A24; 65F45.

1 Introduction

RH-50 reduced the two Hilbert–Schmidt range-resolvent actions left by RH-49 to directional Hardy energies and positive Stein inequalities [5, 6]. This turns a contour problem into an infinite time-domain energy, but it also raises a structural question: can an exact certificate be low rank, scalar, diagonal, or based on a fixed number of noisy steps?

Large-scale Lyapunov solvers often provide accurate low-rank approximations [1, 2, 4]. Accuracy, however, is not the same as a positive Loewner certificate. Exact domination remembers the complete directional cyclic subspace. We first prove the resulting rank obstruction, then show that it does not force a global inverse or fixed-step contraction. Grouping time into blocks permits a horizon $M = M(\sigma, n)$ that grows with the mixing scale. A finite Krylov sum plus a positive tail background then gives a complete certificate.

The numerical outcome is a route selection. The exact finite Gramians become increasingly high-dimensional even though their observed Hardy energies remain moderate. Fixed-rank and scalar shortcuts are therefore poor targets. A growing-horizon block certificate, on the other hand, is extremely sharp on every stored scale.

2 The two directional Hardy pairs

Let $T = G_{2n,\sigma}$ be the fine folded-Gaussian Galerkin matrix and let $A_c = G_{n,\sigma}$ be its exact coarse Haar block. Relative to the adjacent Haar split,

$$T = \begin{pmatrix} A_c & B \\ C & D \end{pmatrix}.$$

Remove the fine and coarse Perron/parity channels:

$$\begin{aligned} Q_f &= I - P_{f,+} - P_{f,-}, & N_f &= T - P_{f,+} - \lambda_{f,-} P_{f,-}, \\ Q_c &= I - P_{c,+} - P_{c,-}, & N_c &= A_c - P_{c,+} - \lambda_{c,-} P_{c,-}. \end{aligned}$$

Let U be the isometric coarse-to-fine embedding and let r exceed both bulk spectral radii. Define

$$A_B = r^{-1} N_f, \quad X_B = \frac{Q_f U B}{\|B\|_{\mathfrak{S}_2}}, \quad Y_B = U^*, \quad (1)$$

$$A_C = r^{-1} N_c^*, \quad X_C = \frac{C^*}{\|C\|_{\mathfrak{S}_2}}, \quad Y_C = Q_c^*. \quad (2)$$

For $s \in \{B, C\}$, put

$$G_s = \sum_{m \geq 0} A_s^m X_s X_s^* (A_s^*)^m.$$

Cyclicity of the finite trace gives

$$\mathcal{E}_B(r)^2 = \text{tr}(Y_B G_B Y_B^*), \quad \mathcal{E}_C(r)^2 = \text{tr}(Y_C G_C Y_C^*). \quad (3)$$

Thus both RH-50 energies have the same abstract form. We now work with a generic stable pair (A, X) .

3 Minimality, cyclic support, and rank

Let $A \in \mathbb{C}^{d \times d}$, $\text{spr}(A) < 1$, and $X \in \mathbb{C}^{d \times p}$. Define

$$G = \sum_{m=0}^{\infty} A^m X X^* (A^*)^m, \quad \mathcal{K}(A, X) = \text{span}\{A^m \text{Ran } X : m \geq 0\}. \quad (4)$$

By Cayley–Hamilton, powers through $d - 1$ suffice in the second formula.

Theorem 3.1 (Minimal Gramian and cyclic support). *The matrix G is the unique Hermitian solution of*

$$G - AGA^* = XX^*.$$

If $H \geq 0$ and

$$H - AHA^* \geq XX^*, \quad (5)$$

then

$$H \geq G, \quad \text{Ran } G = \mathcal{K}(A, X), \quad \text{Ran } X \subseteq \text{Ran } H, \quad A \text{Ran } H \subseteq \text{Ran } H. \quad (6)$$

Consequently

$$\boxed{\text{rank } H \geq \dim \mathcal{K}(A, X) = \text{rank } G.} \quad (7)$$

Proof. Separating the first term of the convergent series gives the Stein equality. If two Hermitian solutions differ by $D = ADA^*$, then $D = A^m D (A^*)^m \rightarrow 0$, so the solution is unique [3, 8].

Iterating (5) gives

$$H \geq \sum_{m=0}^{M-1} A^m XX^* (A^*)^m + A^M H (A^*)^M.$$

Discard the positive remainder and let $M \rightarrow \infty$, obtaining $H \geq G$. Moreover,

$$v^* G v = \sum_{m \geq 0} \|X^* (A^*)^m v\|^2,$$

so $\text{Ker } G = \mathcal{K}(A, X)^\perp$, which proves the range identity.

The inequality $H \geq XX^*$ implies $\text{Ker } H \subseteq \text{Ker } X^*$, hence $\text{Ran } X \subseteq \text{Ran } H$. Also $H \geq AHA^*$. If $v \in \text{Ker } H$, then

$$0 \leq v^* AHA^* v \leq v^* H v = 0,$$

so $A^* v \in \text{Ker } H$. Hence $\text{Ran } H$ is A -invariant and contains the smallest invariant subspace generated by X . This proves (7). $\square$

Corollary 3.2 (Fixed-rank exact-certificate no-go). *If $\dim \mathcal{K}(A_\nu, X_\nu) \rightarrow \infty$ along a family, no exact positive Stein supersolution family can have uniformly bounded rank.*

Remark 3.3. Low-rank Krylov or ADI factors may still approximate G very accurately. The theorem says that once a cyclic direction is omitted, the factor alone cannot be an exact positive certificate. A positive background completion remains possible.

4 The unavoidable background floor

Write $\lambda_1(G) \geq \dots \geq \lambda_d(G) \geq 0$.

Theorem 4.1 (Low-rank plus isotropic floor). *If*

$$H = ZZ^* + \alpha I \geq G, \quad \text{rank } Z \leq k < d, \quad \alpha \geq 0,$$

then

$$\boxed{\alpha \geq \lambda_{k+1}(G).} \quad (8)$$

Proof. Loewner monotonicity and the min-max principle give $\lambda_{k+1}(G) \leq \lambda_{k+1}(H)$. Since ZZ^* has rank at most k , at most k eigenvalues of H exceed α , and $\lambda_{k+1}(H) = \alpha$ [3]. $\square$

For an observation Y ,

$$\text{tr}(YHY^*) = \|YZ\|_{\mathfrak{S}_2}^2 + \alpha \|Y\|_{\mathfrak{S}_2}^2. \quad (9)$$

In the left pair, $Y_B = U^*$ and $\|Y_B\|_{\mathfrak{S}_2}^2 = n$. Therefore even a small scalar floor may have a dimensional trace cost. This is why an anisotropic background is eventually preferable.

5 Growing-horizon block certificates

For $M \geq 1$, put

$$S_M = \sum_{j=0}^{M-1} A^j X X^* (A^*)^j, \quad \mathcal{D}_M(H) = H - A^M H (A^*)^M. \quad (10)$$

Theorem 5.1 (Block Stein domination). *If $H \geq 0$ and*

$$\mathcal{D}_M(H) \geq S_M, \quad (11)$$

then $H \geq G$. The exact Gramian satisfies $\mathcal{D}_M(G) = S_M$ for every M . Every one-step supersolution satisfies (11) for every M .

Proof. Iterating (11) over L blocks yields

$$H \geq \sum_{k=0}^{L-1} A^{kM} S_M (A^*)^{kM} + A^{LM} H (A^*)^{LM} = \sum_{m=0}^{LM-1} A^m X X^* (A^*)^m + A^{LM} H (A^*)^{LM}.$$

Discard the final positive term and let $L \rightarrow \infty$. Grouping the series for G proves equality. Finally,

$$H - A^M H (A^*)^M = \sum_{j=0}^{M-1} A^j (H - A H A^*) (A^*)^j,$$

which proves the last statement. $\square$

The block theorem is not a rank loophole: every block supersolution still dominates G . Its advantage is temporal. Positivity may become easy only after a horizon that increases with the noise-dependent mixing time.

Theorem 5.2 (Finite Krylov sum plus isotropic tail). *Let $P_M = A^M$, $q_M = \|P_M\| < 1$, and define*

$$R_M = P_M S_M P_M^*, \quad \alpha_M = \frac{\|R_M\|}{1 - q_M^2}, \quad H_M = S_M + \alpha_M I. \quad (12)$$

Then $\mathcal{D}_M(H_M) \geq S_M$, hence $H_M \geq G$. For every observation Y ,

$$\boxed{\text{tr}(Y G Y^*) \leq \text{tr}(Y S_M Y^*) + \frac{\|P_M S_M P_M^*\|}{1 - q_M^2} \|Y\|_{\mathfrak{S}_2}^2.} \quad (13)$$

Moreover

$$\alpha_M \leq \frac{q_M^2 \|S_M\|}{1 - q_M^2}.$$

Proof. Since $I - P_M P_M^* \geq (1 - q_M^2)I$ and $\|R_M\| I \geq R_M$,

$$\mathcal{D}_M(H_M) - S_M = \alpha_M (I - P_M P_M^*) - R_M \geq 0.$$

Apply theorem 5.1, pair $H_M \geq G$ with $Y^* Y$, and use $\|R_M\| \leq q_M^2 \|S_M\|$. $\square$

If $\|Y\|_{\mathfrak{S}_2}^2 \asymp n$, the isotropic tail is controlled once $q_M = O(n^{-1/2})$, provided the finite-horizon quantities are bounded. Exponential decay after a transient would make this a logarithmic horizon $M = O(\log n)$. RH-50 rules out a noise-uniform fixed M ; it does not rule out $M = M(\sigma, n) \rightarrow \infty$.

6 Anisotropic completion and conic witnesses

The identity floor is universal but may be expensive. The next theorem isolates what a structured positive background must prove.

Theorem 6.1 (Residual/background completion). *Let $\hat{H} \geq 0$, $W \geq 0$, and*

$$E_M = S_M - \mathcal{D}_M(\hat{H}).$$

If $\mathcal{D}_M(W) \geq \mu I$ for some $\mu > 0$, then

$$H = \hat{H} + \alpha W, \quad \alpha \geq \frac{\max\{0, \lambda_{\max}(E_M)\}}{\mu}$$

is a block supersolution and dominates G .

Proof. Directly,

$$\mathcal{D}_M(H) - S_M = \alpha \mathcal{D}_M(W) - E_M \geq \alpha \mu I - E_M \geq 0.$$

Apply [theorem 5.1](#). □

Here $\hat{H}$ may be a truncated Krylov factor, banded matrix, or multilevel sum. The background W may weight only the sectors carrying the residual, replacing the cost $\alpha \|Y\|_{\mathfrak{S}_2}^2$ by $\alpha \text{tr}(Y W Y^*)$.

Theorem 6.2 (Positive conic dual witness). *Let $W_1, \dots, W_J \geq 0$. Suppose $Z \geq 0$ satisfies*

$$\text{tr}(Z(W_j - A W_j A^*)) \leq 0 \quad (j = 1, \dots, J), \quad \text{tr}(Z X X^*) > 0.$$

Then no nonnegative conic combination

$$H = \sum_{j=1}^J h_j W_j, \quad h_j \geq 0,$$

satisfies $H - A H A^ \geq X X^*$.*

Proof. If it did, positivity would give

$$0 \leq \text{tr}(Z(H - A H A^* - X X^*)) = \sum_j h_j \text{tr}(Z(W_j - A W_j A^*)) - \text{tr}(Z X X^*) < 0,$$

a contradiction. □

Taking $Z = v v^*$ gives the cheap rank-one test

$$v^*(W_j - A W_j A^*)v \leq 0 \quad \text{for every } j, \quad \|X^* v\|^2 > 0.$$

The same theorem holds for block defects with source S_M .

7 Five-scale dense Gramian audit

The experiment uses the row-normalized folded-Gaussian matrices of RH-50 at the smaller dense-solve resolution

$$N\sigma = 5.12, \quad r = 0.85.$$

Both peripheral channels are removed with dense binary64 eigendata. The complete finite Lyapunov equations are then solved through $N = 512$. Here *exact Gramian* means the complete dense finite-matrix solution, not exact arithmetic or interval validation.

For a positive Gramian define

$$r_{\text{eff}}(G) = \frac{(\text{tr } G)^2}{\text{tr}(G^2)},$$

and let r_{99} be the least rank carrying 99% of the trace. The cyclic ranks below are numerical ranks of $[X, AX, \dots, A^9 X]$ at relative singular threshold 10^{-8} .

σ	N	$\mathcal{E}_B$	$\mathcal{E}_C$	r_{eff}^B	r_{99}^B	r_{eff}^C	r_{99}^C	cyclic
0.16	32	0.9040	1.0026	2.40	5	2.82	5	22
0.08	64	1.1626	1.2653	5.09	9	5.54	9	43
0.04	128	1.3338	1.4845	8.62	17	8.42	17	84
0.02	256	1.4096	1.6340	15.51	35	12.33	33	165
0.01	512	1.4681	1.7603	25.66	69	16.30	64	322

Table 1: Dense five-scale Gramian audit. Energies use complete binary64 Lyapunov solutions. The ranks are diagnostics, not validated asymptotic lower bounds.

The energies remain moderate, but state complexity grows rapidly. The left and right 99% ranks are

$$(5, 9, 17, 35, 69), \quad (5, 9, 17, 33, 64),$$

with fitted dimension powers 0.9533 and 0.9231. The terminal cyclic ranks

$$(22, 43, 84, 165, 322)$$

have fitted power 0.9683 and maximum log residual 0.0021. At the finest scale the thresholded cyclic span occupies 62.9% of the state dimension. This is strong evidence against a fixed-rank physical certificate, but a rigorous physical asymptotic no-go still requires an analytic cyclic-dimension lower bound.

7.1 Simple metrics

For the left pair, the minimum eigenvalues of the unforced identity defects $I - A_B A_B^*$ are

$$0.5064, \quad -0.2002, \quad -1.1961, \quad -2.5744, \quad -4.5093.$$

At the four finest levels, a minimum-eigenvector also has positive source overlap. By [theorem 6.2](#), no scalar multiple hI , $h \geq 0$, is a Stein supersolution there.

Simply extracting the diagonal of the exact Gramian also fails. The minimum eigenvalues of

$$\hat{G} - A_B \hat{G} A_B^* - X_B X_B^*, \quad \hat{G} = \text{diag}(G_B),$$

are

$$-0.3545, \quad -0.2874, \quad -0.2213, \quad -0.1675, \quad -0.1232.$$

This does not rule out an independently optimized diagonal metric, much less a banded or multilevel metric.

7.2 Growing horizons

The experiment checks

$$M \in \{1, 2, 4, 8, 12, 16, 24, 32, 48, 64\}$$

and selects the first checkpoint for which

$$\|A^M\|^2 \|Y\|_{\mathfrak{S}_2}^2 \leq 0.25.$$

It then constructs H_M from [theorem 5.2](#).

σ	N	M	q_M^B	q_M^C	left exact/upper	right exact/upper	max excess
0.16	32	4	0.0210	0.0198	0.9040/0.9042	1.0026/1.0029	3.1×10^{-4}
0.08	64	8	0.0305	0.0279	1.1626/1.1632	1.2653/1.2661	6.3×10^{-4}
0.04	128	16	0.0156	0.0156	1.3338/1.3340	1.4845/1.4850	3.5×10^{-4}
0.02	256	24	0.0133	0.0124	1.4096/1.4097	1.6340/1.6344	2.6×10^{-4}
0.01	512	32	0.0069	0.0067	1.4681/1.4681	1.7603/1.7604	6.7×10^{-5}

Table 2: Growing-horizon isotropic block completions. The defects are positive to binary64 precision, not interval-certified.

The selected horizons are

$$4, \quad 8, \quad 16, \quad 24, \quad 32.$$

A linear fit against $\log_2 N$ has slope 7.2, intercept -33.6 , and maximum residual 1.6. This is only logarithmic-looking finite evidence, but it is the behavior required by the block route. The largest relative energy excess is 5.45×10^{-4} on the left and 6.28×10^{-4} on the right. The smallest binary64 block-defect eigenvalue is 1.34×10^{-13} , so an eventual computer-assisted proof must use outward rounding.

Structured Stein certificates: high-dimensional Gramians and a viable growing-horizon closure

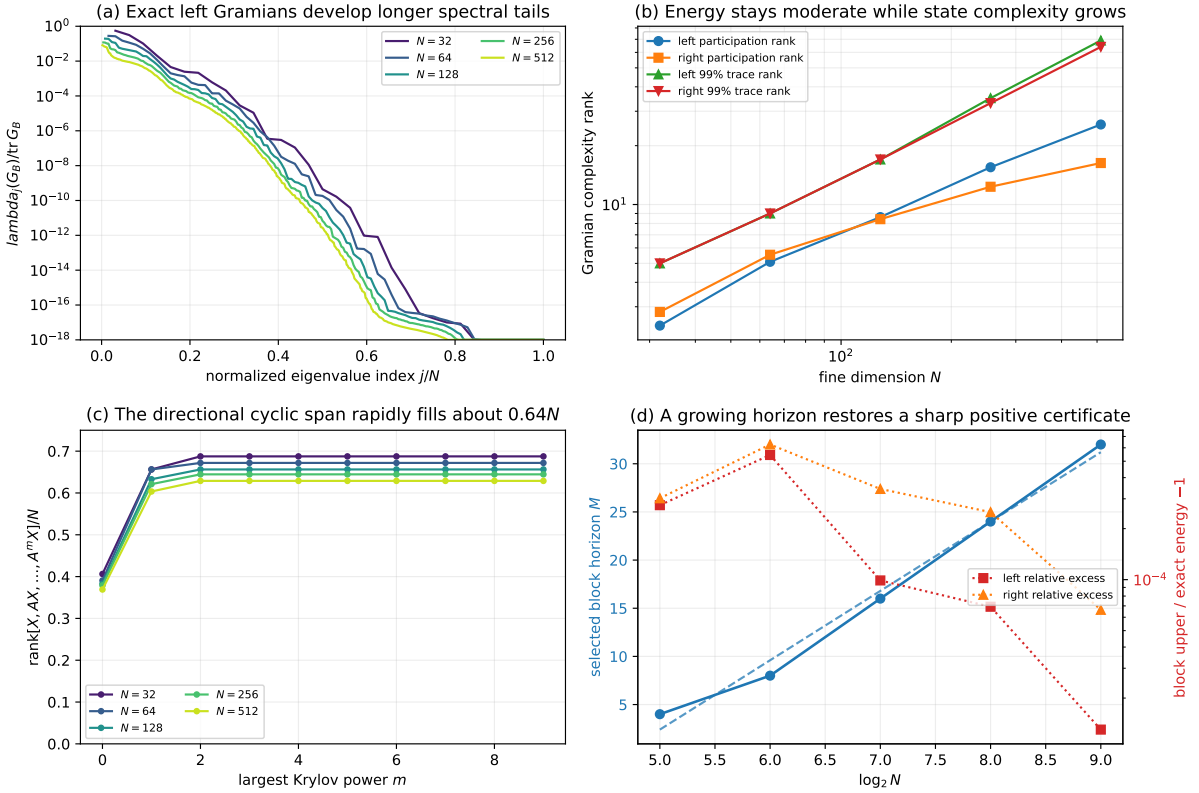


Figure 1: Five-scale audit. (a) The normalized left Gramian spectra develop longer tails. (b) Effective and 99% ranks grow with dimension. (c) The thresholded cyclic span rapidly fills about $0.64N$. (d) A growing horizon gives block energy uppers within 6.3×10^{-4} relative of the dense Gramians.

8 Roadmap consequence and theorem boundary

RH-51 closes several shortcuts and leaves one viable route [7].

Fixed-rank exact factors

Impossible unless the directional cyclic dimension stays uniformly bounded. The physical audit strongly points in the opposite direction.

Scalar identity and extracted diagonal

The identity cone is obstructed at the four finest stored levels, and the diagonal of the exact Gramian fails at all five.

Growing-horizon block route

Viable and numerically sharp. The fixed-step no-go is not a no-go for $M = M(\sigma, n)$.

Stage A1

Not closed. The missing theorem is a dyadically uniform analytic trace budget for an anisotropic or multilevel growing-horizon completion.

The next constructive ansatz should combine a finite Krylov transport part, a positive endpoint/interior multilevel background, and an a posteriori residual bound. Candidate cones should first be screened by [theorem 6.2](#). The scheduled RH-52 peripheral-factor transfer can proceed independently, but the RH-54 intrinsic-identification synthesis still requires this A1 trace budget.

The TPC series remains an independent twin-prime and prime-correlation program. Its methods may be referenced, but it is not an assumption or the arithmetic half of this RH result. We do not construct a prime-power trace formula, identify a zeta zero, build a self-adjoint Hilbert–Pólya operator, derive a $T \log T$ counting law, prove the Riemann hypothesis, or prove a twin-prime statement.

9 Reproducibility

The archive contains the algebra in `src/structured_stein`, the five-scale pilot and certificate builders in `experiments`, the figure, tests, result ledgers, dependency hashes, summary, and archive verification. The principal commands are

```
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/pytest -q
OPENBLAS_NUM_THREADS=16 OMP_NUM_THREADS=16 \
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
    experiments/run_structured_stein_pilot.py
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
    experiments/build_stein_certificate.py
MPLBACKEND=Agg PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
    experiments/make_figures.py
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
    experiments/build_archive.py
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
    experiments/verify_archive.py
```

The minimality, cyclic-support, rank, floor, block, residual-completion, and conic-witness statements are rigorous finite-dimensional theorems. The physical rank growth, logarithmic-looking horizon, simple-metric failures, and near-exact block uppers are binary64 diagnostics at $N\sigma = 5.12$. They are not interval enclosures and do not prove a uniform small-noise Hardy bound.

References

- [1] Athanasios C. Antoulas. *Approximation of Large-Scale Dynamical Systems*, volume 6 of *Advances in Design and Control*. Society for Industrial and Applied Mathematics, Philadelphia, 2005. doi: 10.1137/1.9780898718713.
- [2] Peter Benner, Jing-Rebecca Li, and Thilo Penzl. Numerical solution of large-scale Lyapunov equations, Riccati equations, and linear-quadratic optimal control problems. *Numerical Linear Algebra with Applications*, 15(9):755–777, 2008. doi: 10.1002/nla.622.
- [3] Roger A. Horn and Charles R. Johnson. *Matrix Analysis*. Cambridge University Press, Cambridge, 2 edition, 2013. doi: 10.1017/CBO9781139020411.
- [4] Valeria Simoncini. Computational methods for linear matrix equations. *SIAM Review*, 58(3): 377–441, 2016. doi: 10.1137/130912839.
- [5] Liang Wang. Residue-deflated directional resolvents at a quadratic critical fold: A quarter-power stable-rank bridge for intrinsic Riesz identification. Preprint and software repository, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/RH-49-directional-reduced-resolvent.
- [6] Liang Wang. Two-pole Hilbert–Schmidt hardy energies for small-noise range resolvents: Directional Stein certificates and a global-contraction no-go. Preprint and software repository, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/RH-50-two-pole-hilbert-schmidt-hardy.
- [7] Liang Wang. Quadratic prime-dynamics roadmap after RH-50. Program audit and staged research roadmap, 2026. URL https://github.com/maris205/prime_dynamics_theory/blob/main/papers/RH-ROADMAP-after-RH50.md.
- [8] Kemin Zhou, John C. Doyle, and Keith Glover. *Robust and Optimal Control*. Prentice Hall, Upper Saddle River, NJ, 1996.